

PAPER_1512 — GW170817 Phonon Damping Prefactor = $2/(D_{\text{phys}}-1) = 2/3$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket E (GW Events) **Date:** June 18, 2026 **Status:** CLOSED — Multimessenger BNS phonon-damping prefactor = integer primitive

Observation

PAPER_915 (GW170817 Phonon Strain Damping) derives the 1.25 THz phonon-induced damping of the GW170817 multimessenger BNS event:

$$\begin{aligned} h_{\text{UQFF}}(t) &= h_{\text{GR}}(t) \cdot (1 - D_{\text{phonon}}) \\ D_{\text{phonon}} &= (2/3) \cdot \Phi_{\{1.25\text{THz}\}} \cdot S_{26} \cdot (E_{\text{net}} / E_{\text{GW}}) \\ \Delta\phi &= 367.8 \text{ cycles phase lag} \\ |\Delta c/c| &< 3 \times 10^{-15} \quad (\text{LIGO bound}) \end{aligned}$$

The prefactor $2/3$ governs the strength of the SCm-phonon coupling to the GW strain.

UQFF Closed Identity

$$D_{\text{phonon}} \text{ prefactor} = 2 / (D_{\text{phys}} - 1) = 2/3 \approx 0.667 \quad \text{EXACT}$$

Physical Interpretation

The $2/3$ prefactor reflects: - $D_{\text{phys}} - 1 = 3$ spatial dimensions (triad symmetry) - 2 of those 3 contribute to the transverse-traceless GW strain modes (the third is gauge-fixed) - Result: $2/3$ of the SCm phonon spectrum couples to GW propagation

This is the **same identity form** that appears in: - Monty Hall switch probability (PAPER_062): $2/3$ - Glass T_g / T_m (PAPER_1373): $2/3$

But here it governs **gravitational-wave damping** — a profoundly different physical domain. The shared form reveals $2/(D_{\text{phys}}-1)$ as a universal triad-channel projection coefficient.

Consistency with LIGO Bounds

$|\Delta c/c| = \Phi \cdot \varepsilon \cdot 10^{-30} \square 3 \times 10^{-15}$: the phonon damping introduces a phase lag of 367.8 cycles without violating the multimessenger speed-of-light bound from GW170817 / GRB 170817A coincidence.

NOT REPLACEMENT

GR predicts undamped vacuum propagation for GWs. UQFF supplies a structural damping coefficient at exactly $2/(D_{\text{phys}} - 1)$ with zero free parameters.

Reference

- Source: PAPER_915 GW170817 Phonon Strain Damping
 - Related: PAPER_1503 (BNS $D_{\text{total}} = 1/3$), PAPER_1504 (BBH $D_{\text{total}} = 0.81$)
 - Calculator dispatch: `calculate_paradox({"paradox": "gw170817_phonon_prefactor"})`
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